DBMS Project Task - 6

Project Group Number - 47
Tutorial Group Number - 4
Tutorial Room Number - C210
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Citizen Social Responsibility & Trust Score Management System

- **Type 1 - Lost Update**

Set

```sql
UPDATE citizen SET CurrentScore = 40 WHERE CitizenID = 102;
```

Check

```sql
SELECT CurrentScore FROM citizen WHERE CitizenID = 102;
```

Wrong Behaviour

Terminal 1

```sql
SET autocommit = 0;
START TRANSACTION;

SELECT CurrentScore INTO @score1
FROM citizen
WHERE CitizenID = 102;

DO SLEEP(5);

UPDATE citizen
SET CurrentScore = @score1 + 10
WHERE CitizenID = 102;
COMMIT;
```

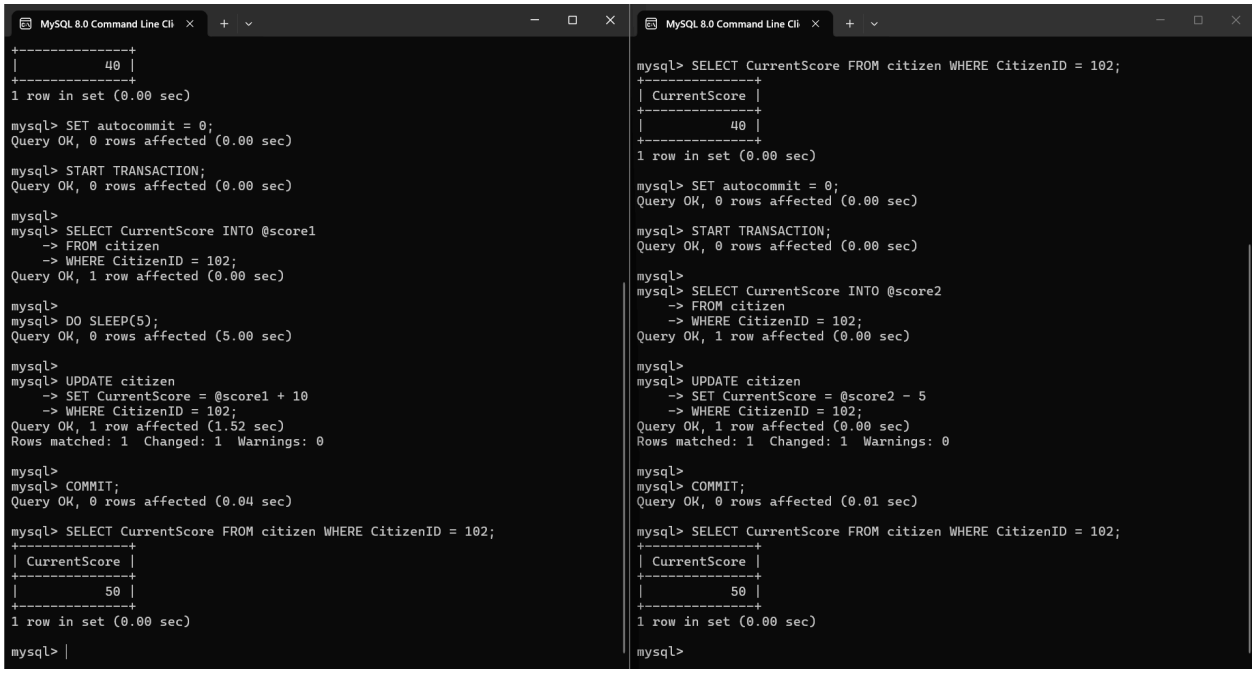
Terminal 2

```sql
SET autocommit = 0;
START TRANSACTION;

SELECT CurrentScore INTO @score2
FROM citizen
WHERE CitizenID = 102;

UPDATE citizen
SET CurrentScore = @score2 - 5
WHERE CitizenID = 102;

COMMIT;
```



Lost Update - Wrong value 50 (Fix using Locking - FOR UPDATE)

Terminal 1

```sql
SET autocommit = 0;
START TRANSACTION;

SELECT CurrentScore
FROM citizen
WHERE CitizenID = 102
FOR UPDATE;
```

```sql
DO SLEEP(5);

UPDATE citizen
SET CurrentScore = CurrentScore + 10
WHERE CitizenID = 102;
COMMIT;
```

Terminal 2

```sql
SET autocommit = 0;
START TRANSACTION;

UPDATE citizen
SET CurrentScore = CurrentScore - 5
WHERE CitizenID = 102;

COMMIT;
```

```
mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT CurrentScore
    -> FROM citizen
    -> WHERE CitizenID = 102
    -> FOR UPDATE;
+--------------+
| CurrentScore |
+--------------+
|           40 |
+--------------+
1 row in set (3.57 sec)

mysql>
mysql> DO SLEEP(5);
Query OK, 0 rows affected (5.00 sec)

mysql>
mysql> UPDATE citizen
    -> SET CurrentScore = CurrentScore + 10
    -> WHERE CitizenID = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT CurrentScore FROM citizen WHERE CitizenID = 102;
+--------------+
| CurrentScore |
+--------------+
|           45 |
+--------------+
1 row in set (0.00 sec)
```

```
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT CurrentScore FROM citizen WHERE CitizenID = 102;
+--------------+
| CurrentScore |
+--------------+
|           50 |
+--------------+
1 row in set (0.00 sec)

mysql> UPDATE citizen SET CurrentScore = 40 WHERE CitizenID = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE citizen
    -> SET CurrentScore = CurrentScore - 5
    -> WHERE CitizenID = 102;
Query OK, 1 row affected (5.06 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT CurrentScore FROM citizen WHERE CitizenID = 102;
+--------------+
| CurrentScore |
+--------------+
|           45 |
+--------------+
1 row in set (0.00 sec)

mysql>
```

- **Type 2 - <u>Dirty Read (Not Possible to show in SQL due to InnoDB handles automatically but theoretically yes)</u>**

Terminal 1

```
SQL
SET autocommit = 0;
START TRANSACTION;

UPDATE citizen
SET CurrentScore = 20
WHERE CitizenID = 103;
```

Terminal 2

```
SQL
SET SESSION TRANSACTION ISOLATION LEVEL READ UNCOMMITTED;

SELECT CurrentScore
FROM citizen
WHERE CitizenID = 103;
```

Then ROLLBACK in Terminal 1
AS in READ UNCOMMITTED Terminal 2 will show uncommitted but updated 20 value which in real is ROLLBACKED in terminal 1 there after resulting in dirty read value;

Theoretically Fix
Terminal 2 - USE COMMITTED ONLY

```
SQL
SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED;

SELECT CurrentScore
FROM citizen
WHERE CitizenID = 103;
```

- **Type 3 - <u>Repeatable Read - Double Read Within same Transaction</u>**

Terminal 1

```
SQL
SET autocommit = 0;
SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED;

START TRANSACTION;
```

```sql
SELECT CurrentScore
FROM citizen
WHERE CitizenID = 104;
```

Terminal 2

```sql
SET autocommit = 0;
START TRANSACTION;

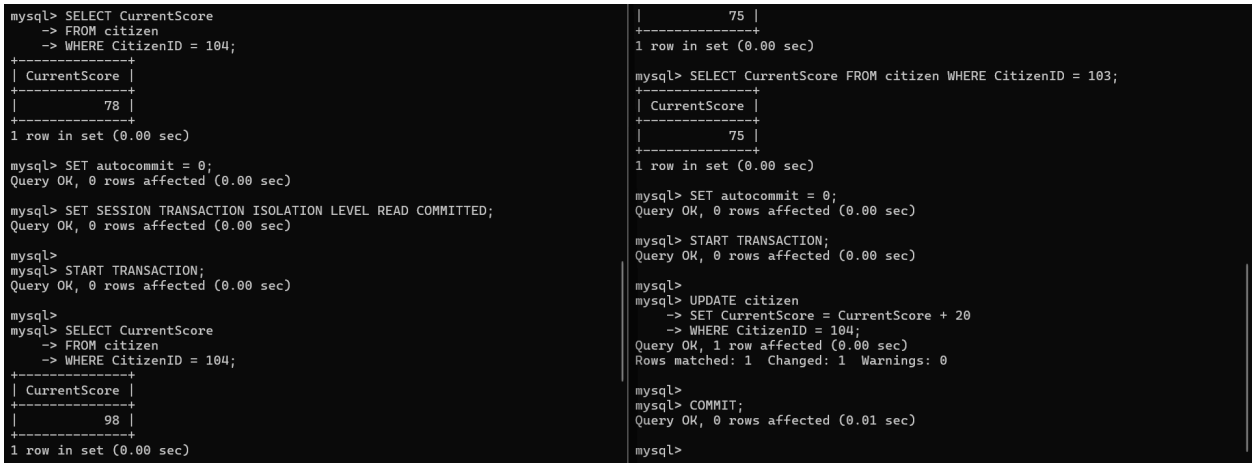
UPDATE citizen
SET CurrentScore = CurrentScore + 20
WHERE CitizenID = 104;

COMMIT;
```

Terminal 1 Again Within Same Transaction

```sql
SELECT CurrentScore
FROM citizen
WHERE CitizenID = 104;
```

Wrong Behaviour is that the value will get updated within the same transaction



Fix

Terminal 1

```sql
SET autocommit = 0;
SET SESSION TRANSACTION ISOLATION LEVEL REPEATABLE READ;
```

```sql
START TRANSACTION;

SELECT CurrentScore
FROM citizen
WHERE CitizenID = 104;
```

Terminal 2

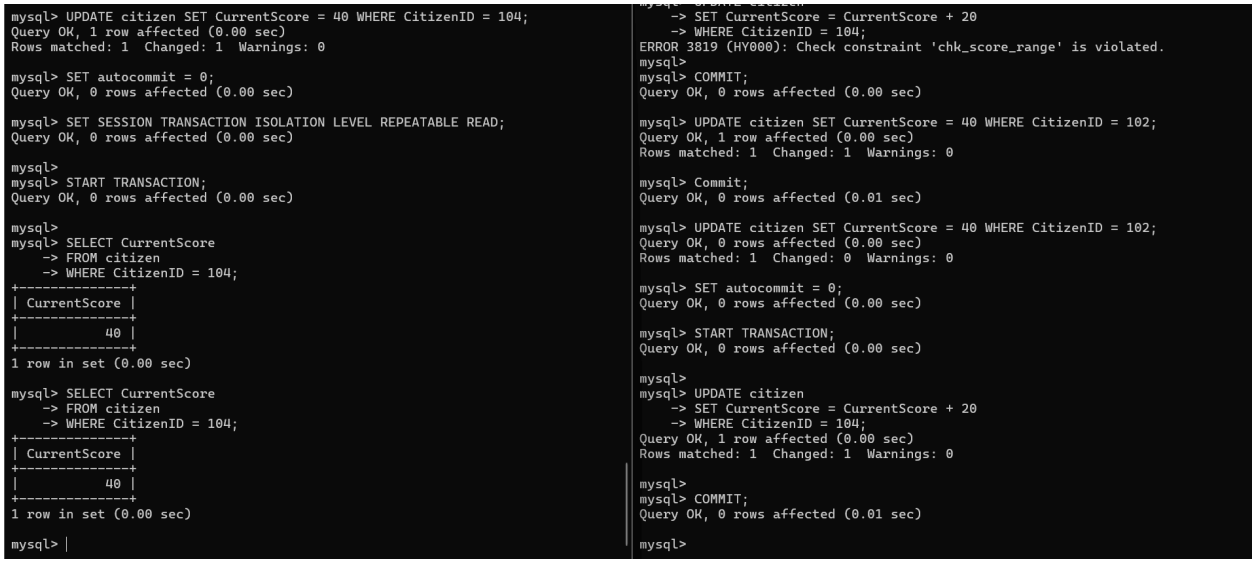
```sql
SET autocommit = 0;
START TRANSACTION;

UPDATE citizen
SET CurrentScore = CurrentScore + 20
WHERE CitizenID = 104;

COMMIT;
```

Terminal 1 Again

```sql
SELECT CurrentScore
FROM citizen
WHERE CitizenID = 104;
```



```
mysql> UPDATE citizen SET CurrentScore = 40 WHERE CitizenID = 104;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.00 sec)

mysql> SET SESSION TRANSACTION ISOLATION LEVEL REPEATABLE READ;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT CurrentScore
    -> FROM citizen
    -> WHERE CitizenID = 104;
+--------------+
| CurrentScore |
+--------------+
|           40 |
+--------------+
1 row in set (0.00 sec)

mysql> SELECT CurrentScore
    -> FROM citizen
    -> WHERE CitizenID = 104;
+--------------+
| CurrentScore |
+--------------+
|           40 |
+--------------+
1 row in set (0.00 sec)

mysql>
```

```
    -> SET CurrentScore = CurrentScore + 20
    -> WHERE CitizenID = 104;
ERROR 3819 (HY000): Check constraint 'chk_score_range' is violated.
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE citizen SET CurrentScore = 40 WHERE CitizenID = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> Commit;
Query OK, 0 rows affected (0.01 sec)

mysql> UPDATE citizen SET CurrentScore = 40 WHERE CitizenID = 102;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0

mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE citizen
    -> SET CurrentScore = CurrentScore + 20
    -> WHERE CitizenID = 104;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql>
```

- **Type 4 - <u>Phantom Read (Insertion/Deletion by some other transaction)</u>**

Terminal 1

```sql
SET autocommit = 0;
SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED;

START TRANSACTION;
SELECT COUNT(*)
FROM complaint
WHERE Status = 'Pending';
DO SLEEP(5);
SELECT COUNT(*)
FROM complaint
WHERE Status = 'Pending';

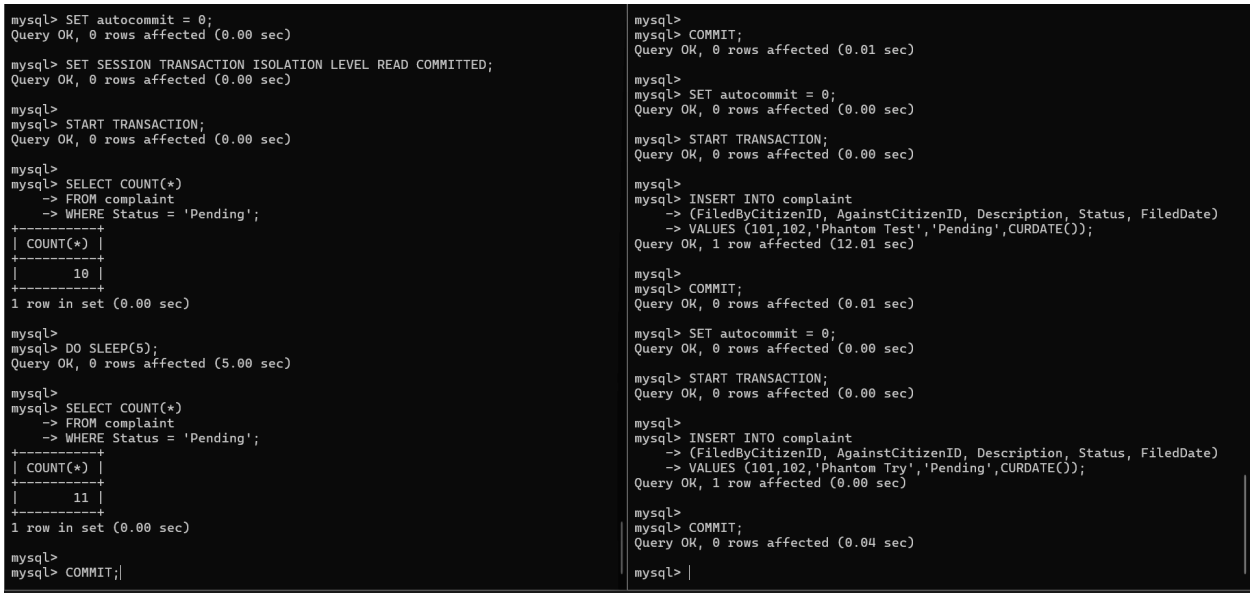
COMMIT;
```

Terminal 2

```sql
SET autocommit = 0;
START TRANSACTION;

INSERT INTO complaint
(FiledByCitizenID, AgainstCitizenID, Description, Status, FiledDate)
VALUES (101,102,'Phantom Try','Pending',CURDATE());

COMMIT;
```



```
mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.00 sec)

mysql> SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT COUNT(*)
    -> FROM complaint
    -> WHERE Status = 'Pending';
+----------+
| COUNT(*) |
+----------+
|       10 |
+----------+
1 row in set (0.00 sec)

mysql>
mysql> DO SLEEP(5);
Query OK, 0 rows affected (5.00 sec)

mysql>
mysql> SELECT COUNT(*)
    -> FROM complaint
    -> WHERE Status = 'Pending';
+----------+
| COUNT(*) |
+----------+
|       11 |
+----------+
1 row in set (0.00 sec)

mysql>
mysql> COMMIT;
```

```
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO complaint
    -> (FiledByCitizenID, AgainstCitizenID, Description, Status, FiledDate)
    -> VALUES (101,102,'Phantom Test','Pending',CURDATE());
Query OK, 1 row affected (12.01 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO complaint
    -> (FiledByCitizenID, AgainstCitizenID, Description, Status, FiledDate)
    -> VALUES (101,102,'Phantom Try','Pending',CURDATE());
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.04 sec)

mysql>
```

Fix with SERIALIZABLE - LOCK INSERTION DELETION UNTIL COMMIT

Terminal 1

```sql
SET autocommit = 0;
SET SESSION TRANSACTION ISOLATION LEVEL SERIALIZABLE;

START TRANSACTION;
SELECT COUNT(*)
FROM complaint
WHERE Status = 'Pending';
DO SLEEP(5);
SELECT COUNT(*)
FROM complaint
WHERE Status = 'Pending';

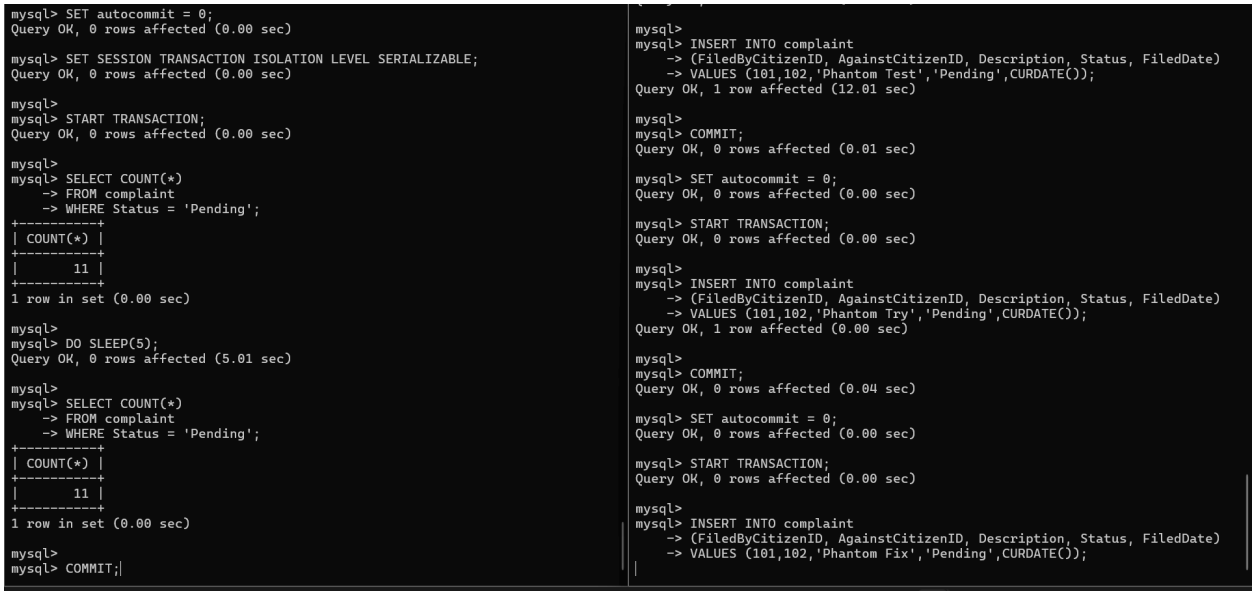
COMMIT;
```

Terminal 2

```sql
SET autocommit = 0;
START TRANSACTION;

INSERT INTO complaint
(FiledByCitizenID, AgainstCitizenID, Description, Status, FiledDate)
VALUES (101,102,'Phantom Fix','Pending',CURDATE());

COMMIT;
```



```
mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.00 sec)

mysql> SET SESSION TRANSACTION ISOLATION LEVEL SERIALIZABLE;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT COUNT(*)
    -> FROM complaint
    -> WHERE Status = 'Pending';
+----------+
| COUNT(*) |
+----------+
|       11 |
+----------+
1 row in set (0.00 sec)

mysql>
mysql> DO SLEEP(5);
Query OK, 0 rows affected (5.01 sec)

mysql>
mysql> SELECT COUNT(*)
    -> FROM complaint
    -> WHERE Status = 'Pending';
+----------+
| COUNT(*) |
+----------+
|       11 |
+----------+
1 row in set (0.00 sec)

mysql>
mysql> COMMIT;
```

```
mysql>
mysql> INSERT INTO complaint
    -> (FiledByCitizenID, AgainstCitizenID, Description, Status, FiledDate)
    -> VALUES (101,102,'Phantom Test','Pending',CURDATE());
Query OK, 1 row affected (12.01 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO complaint
    -> (FiledByCitizenID, AgainstCitizenID, Description, Status, FiledDate)
    -> VALUES (101,102,'Phantom Try','Pending',CURDATE());
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.04 sec)

mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO complaint
    -> (FiledByCitizenID, AgainstCitizenID, Description, Status, FiledDate)
    -> VALUES (101,102,'Phantom Fix','Pending',CURDATE());
```

- **Transaction - Trigger + ROLLBACK**

CHECK

```sql
SELECT CurrentScore FROM citizen WHERE CitizenID = 101;
SELECT * FROM score_history WHERE CitizenID = 101 ORDER BY ChangeDate DESC
LIMIT 1;
```

```
mysql> SELECT CurrentScore FROM citizen WHERE CitizenID = 101;
+--------------+
| CurrentScore |
+--------------+
|           70 |
+--------------+
1 row in set (0.00 sec)

mysql> SELECT * FROM score_history WHERE CitizenID = 101 ORDER BY ChangeDate DESC LIMIT 1;
+-----------+-----------+----------+----------+-------------------+------------+
| HistoryID | CitizenID | OldScore | NewScore | Reason            | ChangeDate |
+-----------+-----------+----------+----------+-------------------+------------+
|        48 |       101 |       50 |       70 | Community Service | 2026-04-09 |
+-----------+-----------+----------+----------+-------------------+------------+
1 row in set (0.00 sec)
```

UPDATE

```sql
SET autocommit = 0;
START TRANSACTION;
INSERT INTO citizen_action_record
(CitizenID, ActionTypeID, OfficialID, ActionDate)
VALUES (101, 3, 1, CURDATE());
SELECT CurrentScore FROM citizen WHERE CitizenID = 101;
SELECT * FROM score_history WHERE CitizenID = 101 ORDER BY ChangeDate DESC
LIMIT 1;
```

```
mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO citizen_action_record
    -> (CitizenID, ActionTypeID, OfficialID, ActionDate)
    -> VALUES (101, 3, 1, CURDATE());
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT CurrentScore FROM citizen WHERE CitizenID = 101;
+--------------+
| CurrentScore |
+--------------+
|           90 |
+--------------+
1 row in set (0.00 sec)

mysql> SELECT * FROM score_history WHERE CitizenID = 101 ORDER BY ChangeDate DESC LIMIT 1;
+-----------+-----------+----------+----------+-------------------+------------+
| HistoryID | CitizenID | OldScore | NewScore | Reason            | ChangeDate |
+-----------+-----------+----------+----------+-------------------+------------+
|        49 |       101 |       70 |       90 | Community Service | 2026-04-09 |
+-----------+-----------+----------+----------+-------------------+------------+
```

```sql
ROLLBACK;
```

```
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT CurrentScore FROM citizen WHERE CitizenID = 101;
+--------------+
| CurrentScore |
+--------------+
|           70 |
+--------------+
1 row in set (0.00 sec)

mysql> SELECT * FROM score_history WHERE CitizenID = 101 ORDER BY ChangeDate DESC LIMIT 1;
+-----------+-----------+----------+----------+------------------+------------+
| HistoryID | CitizenID | OldScore | NewScore | Reason           | ChangeDate |
+-----------+-----------+----------+----------+------------------+------------+
|        48 |       101 |       50 |       70 | Community Service | 2026-04-09 |
+-----------+-----------+----------+----------+------------------+------------+
1 row in set (0.00 sec)
```

- **Share IN LOCK (CAN READ but NOT WRITE)**

Terminal 1

```sql
SET autocommit = 0;
START TRANSACTION;

SELECT CurrentScore
FROM citizen
WHERE CitizenID = 101
LOCK IN SHARE MODE;

DO SLEEP(5);
```

Terminal 2

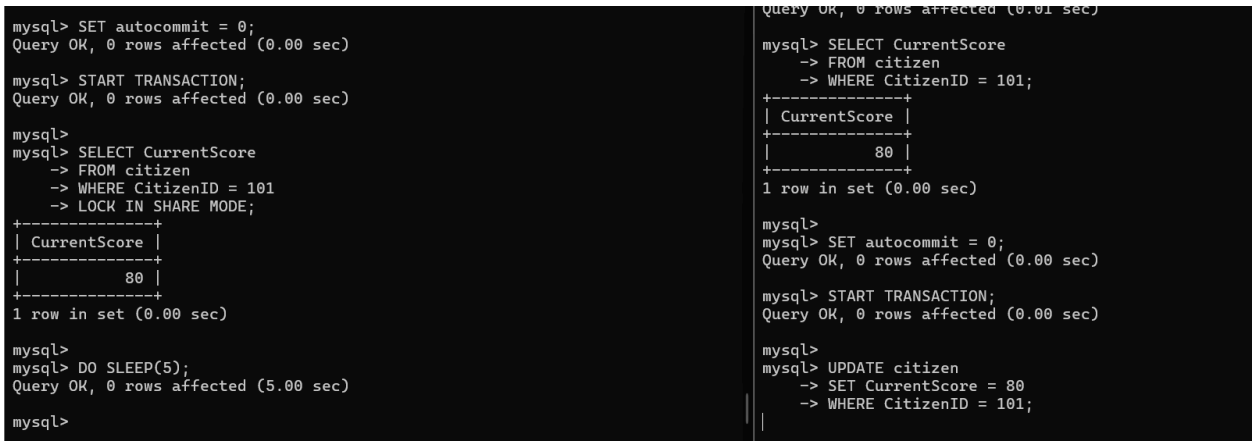
```sql
SELECT CurrentScore
FROM citizen
WHERE CitizenID = 101;

SET autocommit = 0;
START TRANSACTION;

UPDATE citizen
SET CurrentScore = 80
```

```
WHERE CitizenID = 101;

COMMIT;
```



Waiting for T1 to commit - Read Done but Write Blocked

- **SavePoints in Transactions**

```SQL
UPDATE citizen SET CurrentScore = 50 WHERE CitizenID = 101;
SELECT CurrentScore FROM citizen WHERE CitizenID = 101;

SET autocommit = 0;
START TRANSACTION;
UPDATE citizen
SET CurrentScore = 60
WHERE CitizenID = 101;

SAVEPOINT sp1;

UPDATE citizen
SET CurrentScore = 150
WHERE CitizenID = 101;
SELECT CurrentScore FROM citizen WHERE CitizenID = 101;

ROLLBACK TO sp1;

SELECT CurrentScore FROM citizen WHERE CitizenID = 101;
COMMIT;
SELECT CurrentScore FROM citizen WHERE CitizenID = 101;
```

```
mysql> UPDATE citizen SET CurrentScore = 50 WHERE CitizenID = 106;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT CurrentScore FROM citizen WHERE CitizenID = 106;
+--------------+
| CurrentScore |
+--------------+
|           50 |
+--------------+
1 row in set (0.00 sec)
```

```
mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> INSERT INTO citizen_action_record
    -> (CitizenID, ActionTypeID, OfficialID, ActionDate)
    -> VALUES (106, 3, 1, CURDATE());
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT CurrentScore FROM citizen WHERE CitizenID = 106;
+--------------+
| CurrentScore |
+--------------+
|           70 |
+--------------+
1 row in set (0.00 sec)

mysql>
mysql> SAVEPOINT sp1;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> INSERT INTO citizen_action_record
    -> (CitizenID, ActionTypeID, OfficialID, ActionDate)
    -> VALUES (106, 1, 1, CURDATE());
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT CurrentScore FROM citizen WHERE CitizenID = 106;
+--------------+
| CurrentScore |
+--------------+
|           69 |
+--------------+
1 row in set (0.00 sec)

mysql>
mysql> ROLLBACK TO sp1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT CurrentScore FROM citizen WHERE CitizenID = 106;
+--------------+
| CurrentScore |
+--------------+
|           70 |
+--------------+
1 row in set (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> SELECT CurrentScore FROM citizen WHERE CitizenID = 106;
+--------------+
| CurrentScore |
+--------------+
|           70 |
+--------------+
1 row in set (0.00 sec)
```